

WTI-Brent-Dated-MEH Decomposition Engine

Architecture Discussion Draft — for review & correction

Prepared for study · v0.1 working draft · June 2026

What this is. A draft of how I currently understand the engine you want to build — an export-arbitrage *anomaly monitor* over the WTI / Brent / Dated Brent / MEH complex that detects when the relationship is offside, *attributes* the dislocation to a specific component, and proposes a trade with calibrated conviction. **This is deliberately a conversation starter, not a final spec.** Section 9 is the important part: a set of questions about how the relationships actually work, for you to correct tomorrow. Where I've made an assumption about market mechanics, I've flagged it so you can overwrite it.

1 · Mission

A tool that continuously watches the benchmark complex and flags **dislocations** — moments when WTI, Brent, Dated Brent and MEH have drifted out of line with what the US-export arbitrage says they should be. When it flags one, it must answer *exactly where the dislocation lives* (the WTI time structure? the MEH export gateway? freight? the WTI/Brent leg? the Dated structure?), recommend the cleanest way to express it (a single component, or the basket), size it, and attach an *earned* conviction score. Your edge is your understanding of the mechanism; the engine's job is surveillance, decomposition, and discipline at a scale the eye can't cover.

2 · The benchmark complex

The four benchmarks are linked by one physical fact: a barrel of US light sweet crude only reaches the international (Brent) pricing world by leaving the US Gulf as a waterborne cargo. **MEH (Argus WTI Houston) is the node where the domestic WTI barrel becomes an exportable, FOB barrel** — it is the hinge of the entire complex. Cushing WTI is landlocked; you cannot export from Oklahoma.

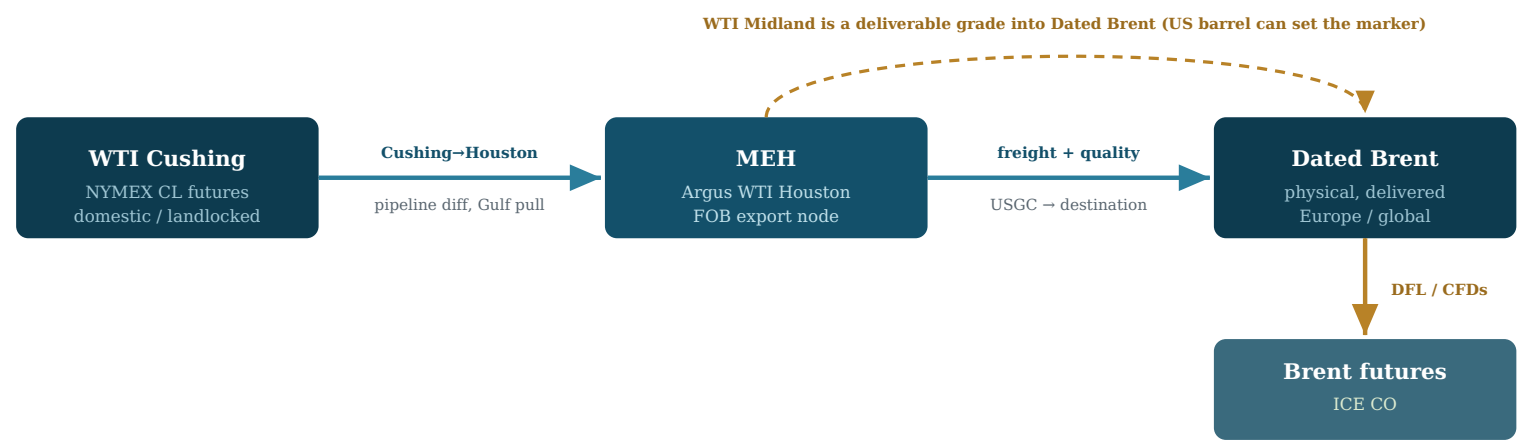


Figure 1 · The export chain. Solid teal = the physical arb path (origin→destination). Gold = the Dated-vs-futures structural links and the WTI-Midland-into-Dated mechanism. *Flag for your confirmation: the WTI-Midland-in-Dated linkage and how tightly it binds the two ends.*

Assumption I need you to confirm (Section 9, Q2): because WTI Midland is now a deliverable grade in the Dated Brent basket, the US barrel frequently *sets* Dated rather than merely competing with it. If true, large parts of the complex are near-arbitraged most of the time, and the tradable anomalies are the *deviations* from that tight linkage. If you see Dated as more independent, the engine's fair-value layer is built differently.

3 · The core arbitrage identity

With MEH at the centre, the export arb that should hold (approximately, net of frictions) is:

$$\text{Dated Brent} \approx \text{WTI Cushing} + (\text{MEH} - \text{Cushing}) + \text{Freight} + \text{Quality}$$

i.e. delivered US barrel value vs the European marker

Rearranged, the **export arb** = Dated – WTI – (MEH – Cushing) – Freight – Quality. When that sits outside its normal band, the complex is offside. The engine's first job is to measure that gap; its second is to decompose it.

4 · The additive decomposition (the keystone)

The reason detection and trade-idea generation are *one* operation: the total dislocation decomposes, additively, into the dislocation of each component — and each component is itself observable and tradable. When the complex is, say, +2σ rich, the same decomposition that detected it tells you *which legs own how much of it*.

Component	What it is	Primary drivers	Traded as
WTI time structure	Shape of the WTI Cushing curve (the CM spreads you've already built)	Cushing storage, prompt US supply/demand, roll dynamics	CL calendar spreads

Cushing→Houston (MEH diff)	The export-gateway differential; cost/availability of getting barrels to water	Pipeline capacity, Gulf export pull, dock/terminal constraints	MEH—Cushing diff (Argus)
WTI/Brent (flat)	Outright transatlantic value gap, the headline arb	Relative supply/demand, global vs US balances	CL—CO futures spread
Freight	Cost to move USGC→destination cargo	Vessel supply, tonne-mile demand, bunker, congestion	Freight (Baltic / WS routes)
Dated structure	Dated-vs-futures basis — prompt physical tightness	North Sea loading programmes, refinery pull, arb-in barrels	DFL / Brent CFDs
Quality	Light-sweet grade adjustment WTI vs the Brent basket	Light/heavy & sweet/sour diffs, product cracks	(usually an input, sometimes a grade-diff trade)

The sum of the component dislocations $\approx$ the total complex dislocation. Additivity is approximate (the legs interact, there is basis risk, attribution carries noise) but it is a strong enough organising principle to build the whole engine around.

5 · Engine architecture (four layers)

Layer	Function
A. Fair value & detection	Compute the arb identity's "fair" relationship from inputs (price + MEH diff + freight + quality, lag-aligned), measure the live gap, and score how offside it is — statistically (vs its own history) and economically (is the arb open/closed).
B. Attribution	Decompose the total dislocation into the six components in §4. Output: how many σ (and \$) each leg owns. This is the "tell me exactly why."
C. Conviction (confluence)	Score conviction from <i>agreement across legs and corroborating fundamentals</i> , calibrated to historical reliability — not from the magnitude of any single z. (See §6.)
D. Expression, sizing & risk	Decide component-vs-basket from the attribution shape, then size and risk-cap using the harness you've already built. Output: a trade-idea ticket.

Layers B–D reuse the constant-maturity construction, backtest simulator, vol-target sizing, risk rails, and calibration mechanism already built and verified in the WTI-spread engine. None of that work is discarded; it becomes the "WTI time structure" component and the shared plumbing.

6 · Why conviction = confluence, not magnitude

The WTI-spread calibration surfaced a real lesson: because a z-rule fires the moment z crosses its threshold, the trigger-z barely varies, so z-magnitude collapses to a single value and can't form a conviction *gradient*. The decomposition engine fixes this by sourcing conviction from a different axis: **when independent legs of the decomposition point the same way, and fundamentals corroborate, conviction is high**. A dislocation that shows up *consistently* across WTI/Brent, the MEH diff, and freight — and lines up with inventories/flows — is far more trustworthy than a single 3σ print on one leg. Conviction becomes multi-dimensional and empirically calibrated, which is exactly what "utmost conviction" should mean: not certainty (no model offers that), but a high, *justified*, honestly-scored probability — and an equally honest "sit this one out" when the legs disagree.

7 · Data architecture & sourcing

The engine needs more than price. The gating inputs — the ones that decide whether this is buildable on your data access — are MEH, freight, and the Platts Dated/CFD assessments. Tags below are my best understanding; confirm against your actual entitlements.

Input	Source	Access	Note
WTI Cushing / CL curve	CME / Bloomberg	EASY	Already in hand.
Brent futures (CO)	ICE / Bloomberg	EASY	Straightforward.
MEH (Argus WTI Houston)	Argus	PAID	Subscription assessment — the key gating input. Hinge of the whole model.
Dated Brent / CFDs / DFL	Platts (S&P Global)	PAID	Physical assessments; licensing matters.
Freight (USGC routes)	Baltic Exchange / broker	PAID	Route-level WS\$/bbl. Some lagged free proxies exist.
Inventories / flows (corroboration)	EIA (US); Kpler/Vortexa (waterborne)	MIXED	EIA free; cargo-tracking flows are paid.
Positioning	CFTC COT	FREE	Weekly, optional lens.

8 · Questions for you — how the relationships actually work

This is the part I most need. Mark these up however you like; your answers determine how sophisticated and how *correct* the engine is. Don't worry about precision of wording — concepts are what matter.

- Q1 Which arb is the anchor in your head?** When you watch the complex, is the primary reference the **WTI/Brent futures spread** (CL-CO), the **physical MEH-Dated** delivered arb, or the **full export arb** (Dated – MEH – freight – quality)? Which one is "truth" and which are derived from it?
- Q2 How tightly does WTI Midland set Dated?** Given WTI Midland is a deliverable Dated grade, do you treat Dated as effectively "US barrel + freight + quality" much of the time (tightly arbitrated), or as genuinely independent North Sea pricing that US *competes against*? This is the single biggest fork in how I build the fair-value layer.
- Q3 Quality — constant or dynamic?** How do you handle the WTI-Brent quality differential, and the WTI-Midland-into-Dated quality premium? A fixed assumption, or something you track dynamically (and from where)? What actually moves it that we'd need to capture?
- Q4 Which freight routes and vessels matter?** USGC→Rotterdam/Augusta on Aframax/Suezmax? USGC→Asia on VLCC? Is freight a *traded leg* for you, or purely an input to the arb calc? Which assessment(s) do you anchor to?
- Q5 Transit lag — how should we time-align?** A cargo lifted at MEH today prices into Europe weeks later. Should the engine compare **spot MEH vs forward Dated** (lag-aligned to voyage time), or spot-vs-spot? How do *you* line them up when you assess whether the arb is open?
- Q6 Are DFL / CFDs their own signal?** Do you watch the Dated-vs-futures structure as an independent read on prompt physical tightness, and should it be a first-class component of the decomposition (vs folded into "Dated")?
- Q7 Scope — Atlantic only, or East/West too?** Is this purely US→Europe, or do you also weigh the Brent/Dubai EFS (US/Atlantic→Asia) as the competing destination for the same barrel? Including it lets the engine reason about *where the marginal barrel goes*.
- Q8 What does "offsides" mean to you?** Primarily (i) statistically stretched vs recent history, (ii) the delivered economics crossing open/closed, or (iii) divergence from a structural fair-value model? Likely a blend — but which fires *first* for you as the alert?
- Q9 How fast do these dislocations resolve?** Typical time for an arb dislocation to correct — days, weeks? This sets the holding period, the backtest horizon, and the lookbacks.
- Q10 Component vs basket — what decides?** You said "both, depending on attribution." Is the logic "concentrated in one leg → trade that leg; diffuse → trade the basket," or do liquidity / execution / margin realities drive the choice more than the attribution shape?

9 · Suggested path (for your reaction)

- **Step 1:** You mark up Sections 2–4 and answer Section 8. We lock the relationship model together.
- **Step 2:** I rewrite this into a build spec (like the backtest spec) — the version you hand Claude Code.
- **Step 3:** First vertical slice end-to-end on the cleanest arb you have data for (likely WTI/Brent flat + MEH diff, since freight/Platts may need sourcing). Detect → attribute → conviction → ticket.
- **Step 4:** Add legs as data is sourced (freight, Dated structure, fundamentals), each plugging into the same decomposition.

Working draft for discussion. Market-structure details herein reflect my current understanding and are flagged for your confirmation; nothing here is a final design. No claim is made that the legacy WTI-spread signal has a tradable edge — the decomposition engine is a distinct, more targeted instrument.